

International Conference on Intelligent, Resilient, and Sustainable Energy Systems (IRSES 2027)

Track 1: AI and Data-Driven Energy Systems

- Machine learning for energy forecasting, optimization, and control
- Foundation models and multimodal learning for energy data
- Explainable and trustworthy AI in energy systems
- AI for Energy System
- Causal inference and uncertainty quantification

Track 2: Cyber-Physical and IoT-Enabled Energy Infrastructure

- Smart grids and cyber-physical energy systems
- IoT architectures for sensing, monitoring, and control
- Edge computing and real-time energy analytics
- Digital twins for energy systems
- Security and privacy in energy infrastructure

Track 3: Resilient, Decentralized, and Market-Driven Energy Systems

- Grid resilience under climate and extreme events
- Distributed energy resources (DERs), microgrids
- Demand response and transactive energy systems
- Energy markets, pricing, and game-theoretic models
- Policy-aware and regulation-driven system design

Track 4: Sustainable Energy Technologies and Circular Systems (Green and Clean Energy)

- Renewable energy systems (solar, wind, hydrogen, and hybrid systems)
- Energy storage technologies and optimization
- Waste-to-energy and circular energy systems
- Lifecycle analysis and carbon-aware design
- Integrated energy systems (electric + thermal + hydrogen)
- 2D Materials for Photovoltaic
- AI for Climate Control

Track 5: Energy Efficient & Secure Hardware Architecture for Edge AI System

- Quantum-inspired and next-gen hardware
- In-memory computing for AI
- Real-time edge intelligence systems
- Secure hardware design for AI systems
- GPU/NPU optimization for edge inference
- Low-power VLSI architectures for AI acceleration
- Energy-aware processor design for edge devices
- Neuromorphic and bio-inspired architectures
- FPGA-based AI accelerators for edge computing